

FULL DETAILS (Read-only) -> [Click Here to Create PDF for Current Dataset of Trial](#)

CTRI Number	CTRI/2024/01/061076 [Registered on: 02/01/2024] Trial Registered Prospectively															
Last Modified On:	11/08/2025															
Post Graduate Thesis	No															
Type of Trial	Observational															
Type of Study	Cohort Study															
Study Design	Single Arm Study															
Public Title of Study	A study to observe the journey of patients undergoing cancer surgeries having higher than average risk.															
Scientific Title of Study	Perioperative Journey of Patients Undergoing Elevated Risk Cancer Surgery - A Prospective Observational Study (POJERC Study)															
Trial Acronym	POJERC Study															
Secondary IDs if Any	<table><tr><th>Secondary ID</th><th>Identifier</th></tr><tr><td>NIL</td><td>NIL</td></tr></table>	Secondary ID	Identifier	NIL	NIL											
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NIL	NIL															
Details of Principal Investigator or overall Trial Coordinator (multi-center study)	<table><tr><td>Name</td><td>Dr Uddalak Chattopadhyay</td></tr><tr><td>Designation</td><td>Associate Professor</td></tr><tr><td>Affiliation</td><td>Mahamana Pandit Madan Mohan Malviya Cancer Centre and Homi Bhabha Cancer Hospital , Varanasi</td></tr><tr><td>Address</td><td>Department of Anaesthesiology Critical Care and Pain, 4th Floor OT Complex, Mahamana Pandit Madan Mohan Malviya Cancer Centre (MPMMCC), Varanasi 221005 Uttar Pradesh India Varanasi UTTAR PRADESH 221005 India</td></tr><tr><td>Phone</td><td>8617279293</td></tr><tr><td>Fax</td><td></td></tr><tr><td>Email</td><td>uddalak@mpmmcc.tmc.gov.in</td></tr></table>		Name	Dr Uddalak Chattopadhyay	Designation	Associate Professor	Affiliation	Mahamana Pandit Madan Mohan Malviya Cancer Centre and Homi Bhabha Cancer Hospital , Varanasi	Address	Department of Anaesthesiology Critical Care and Pain, 4th Floor OT Complex, Mahamana Pandit Madan Mohan Malviya Cancer Centre (MPMMCC), Varanasi 221005 Uttar Pradesh India Varanasi UTTAR PRADESH 221005 India	Phone	8617279293	Fax		Email	uddalak@mpmmcc.tmc.gov.in
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Details of Contact Person Scientific Query	<table><tr><td>Name</td><td>Dr Uddalak Chattopadhyay</td></tr><tr><td>Designation</td><td>Associate Professor ,Department of Anaesthesiology Critical care and Pain</td></tr><tr><td>Affiliation</td><td>MPMMCC & HBCH, Varanasi</td></tr><tr><td>Address</td><td>Department of Anaesthesiology Critical Care and Pain, 4th Floor OT Complex, Mahamana Pandit Madan Mohan Malviya Cancer Centre (MPMMCC), Varanasi 221005 Uttar Pradesh India UTTAR PRADESH 221005 India</td></tr><tr><td>Phone</td><td>8617279293</td></tr><tr><td>Fax</td><td></td></tr><tr><td>Email</td><td>uddalak@mpmmcc.tmc.gov.in</td></tr></table>		Name	Dr Uddalak Chattopadhyay	Designation	Associate Professor ,Department of Anaesthesiology Critical care and Pain	Affiliation	MPMMCC & HBCH, Varanasi	Address	Department of Anaesthesiology Critical Care and Pain, 4th Floor OT Complex, Mahamana Pandit Madan Mohan Malviya Cancer Centre (MPMMCC), Varanasi 221005 Uttar Pradesh India UTTAR PRADESH 221005 India	Phone	8617279293	Fax		Email	uddalak@mpmmcc.tmc.gov.in
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	Phone	8617279293		
	Fax			
	Email	uddalak@mpmmcc.tmc.gov.in		
Source of Monetary or Material Support	Mahamana Pandit Madan Mohan Malviya Cancer Centre and Homi Bhabha Cancer Hospital (MPMMCC and HBCH), Varanasi 221005 Uttar Pradesh India			
Primary Sponsor	Name	Investigator initiated studyDr Uddalak Chattopadhyay		
	Address	Department of Anaesthesiology Critical Care and Pain, 4th Floor OT Complex, Mahamana Pandit Madan Mohan Malviya Cancer Centre (MPMMCC), Varanasi 221005 Uttar Pradesh India		
	Type of Sponsor	Other [Self]		
Details of Secondary Sponsor	Name	Address		
	NIL	NIL		
Countries of Recruitment	India			
Sites of Study	No of Sites = 1			
	Name of Principal Investigator	Name of Site	Site Address	Phone/Fax/Email
	Dr Uddalak Chattopadhyay	MPMMCC VARANASI	Department of Anaesthesiology Critical Care and Pain, 4th Floor OT Complex, Mahamana Pandit Madan Mohan Malviya Cancer Centre (MPMMCC), Varanasi 221005 Uttar Pradesh India Varanasi UTTAR PRADESH	8617279293 uddalak@mpmmcc.tmc.gov.in
Details of Ethics Committee	No of Ethics Committees= 1			
	Name of Committee			Approval Status
	IEC MPMMCC HBCH TMC Varanasi			Approved

Regulatory Clearance Status from DCGI	Status		
	Not Applicable		
Health Condition / Problems Studied	Health Type	Condition	
	Patients	(1) ICD-10 Condition: C768 Malignant neoplasm of other specified ill-defined sites,	
Intervention / Comparator Agent	Type	Name	Details
	Intervention	NIL	Nil
	Comparator Agent	NIL	NIL
Inclusion Criteria	Age From	18.00 Year(s)	
	Age To	90.00 Year(s)	
	Gender	Both	
	Details	Patients aged 18 years and above. Patients scheduled to undergo cancer surgery with an elevated risk for any complication as per the ACS risk calculator.	
ExclusionCriteria	Details	Patients not willing to give consent Cases where proper documented data is not available. Paediatric patients (below 18 years of age) Pregnant patients.	
Method of Generating Random Sequence	Not Applicable		
Method of Concealment	Not Applicable		
Blinding/Masking	Not Applicable		
Primary Outcome	Outcome	TimePoints	
	To validate the American College of Surgeons (ACS) risk calculator in patients undergoing elevated risk cancer surgery. This validation aims to assess the accuracy and applicability of the ACS risk score in predicting postoperative outcomes in the context of high-risk oncology patients.	Intraoperative and postoperative till 30 days of completion of surgery	
Secondary Outcome	Outcome	TimePoints	
	To observe the practice of some of the well-established intraoperative management guidelines on the patients undergoing elevated risk cancer surgery in order to observe their impact on postoperative outcomes: 1. Duration of surgery	Intraoperative and postoperative till 30 days of completion of surgery	

Target Sample Size	Total Sample Size="1000" Sample Size from India="1000" Final Enrollment numbers achieved (Total)= "Applicable only for Completed/Terminated trials" Final Enrollment numbers achieved (India)= "Applicable only for Completed/Terminated trials"
Phase of Trial	N/A
Date of First Enrollment (India)	08/01/2024
Date of Study Completion (India)	Applicable only for Completed/Terminated trials
Date of First Enrollment (Global)	Date Missing
Date of Study Completion (Global)	Applicable only for Completed/Terminated trials
Estimated Duration of Trial	Years="3" Months="0" Days="0"
Recruitment Status of Trial (Global) Modification(s)	Not Applicable
Recruitment Status of Trial (India)	Other (Terminated)
Publication Details	N/A
Individual Participant Data (IPD) Sharing Statement	Will individual participant data (IPD) be shared publicly (including data dictionaries)? Response - NO
Brief Summary	Cancer rates are on the rise globally, posing a significant challenge to public health. The Medical Certification of Cause of Death, 2018 reported cancer as the fifth leading cause of death in India, accounting for 5.7% of all deaths. In the early stages of these cancers, surgery is one of the most effective treatment. Despite the benefits of surgery, however, they are not free of complications. Postoperative complications are frequent events, particularly among patients at high risk. The American College of Surgeons National Surgical Quality Improvement Program (ACS NSQIP) surgical risk calculator (http:// riskcalculator.facs.org) may potentially be used in daily practice, as this online surgical risk calculator originated from a large ACS NSQIP dataset that consisted of 1,414,006 patients who received surgery (of all surgical specialties) between 2009 and 2012 in 393 hospitals in the United States of America.

It predicts the risk for 13 different surgical outcomes (1. Serious complication, 2. Any complication, 3. Pneumonia, 4. Cardiac complication 5. Surgical site infection, 6. Urinary tract infection, 7. Venous Thromboembolism, 8. Renal failure, 9. Readmission, 10. Return to OR, 11. Death, 12. Discharge to Nursing/Rehab facility, 13. Sepsis) based on 21 variables (1. Age, 2. Sex, 3. Functional Status, 4. Nature of procedure – Elective/Emergency, 5. ASA class, 6. History of Steroid use for chronic condition, 7. Presence of Ascites within 30 days prior to surgery, 8. Presence of systemic sepsis within 48 hrs prior to Surgery, 9. Status of ventilator dependency, 10. Presence of disseminated cancer, 11. History of Diabetes Mellitus, 12. History of Hypertension requiring medication, 13. History of congestive heart failure in 30 days prior to surgery, 14. Presence of Dyspnoea, 15. Status of Smoking within one year prior to surgery, 16. History of severe COPD, 17. History of Dialysis within 2 weeks prior to surgery, 18. History of acute renal failure 19. Height, 20. Weight, 21. Surgical procedure). Moreover, the Risk Calculator's risk-estimating methodology has now transitioned from regression to machine learning from July 2023. This change in methodology will improve the calculator's already excellent accuracy, though differences in estimated risk should not be large.⁶ The ACS NSQIP calculator has already been validated in different population cohorts and those studies found a good overall performance.

In addition to preoperative risk factors, intraoperative management can indeed have an effect on postoperative outcomes in cancer patients.

Our study aims to observe the practice of some of the well-established intraoperative management guidelines (stated below) on the patients undergoing elevated risk cancer surgery in order to observe their impact on postoperative outcomes.

Available data suggest strong association between desaturation events (oxygen saturation < 90%) within 5 min of tracheal intubation and adverse outcome in the form of postoperative respiratory complications.

After induction of general endotracheal anaesthesia, lung-protective mechanical ventilation seeks to avoid derecruitment without overdistension of alveoli by keeping transpulmonary pressures such that the lung tissue stays in the linear part of its local pressure–volume curve. Accordingly, the absence of

lung protective ventilation patterns was associated with an increased risk of early tracheal re-intubation.

Although high FiO₂ may improve tissue oxygenation, it can impair pulmonary function. There is evidence of a strong association between high median FIO₂ (> 0.61) and early tracheal re-intubation after surgery.

Duration of surgery also has strong correlation with postoperative adverse outcome.

Increased average minimum alveolar concentration of inhalational anaesthetics (volatile anaesthetics and nitrous oxide) is dose-dependently associated with a lower 30-day mortality, lower hospital cost, and lower risk of PRCs.

The use of intermediate-acting nondepolarizing neuromuscular blocking agents (NMBAs) is associated with an increased risk of postoperative desaturation to SaO₂ less than 90% after extubation and reintubation requiring unplanned admission to an ICU. These same outcomes are increased with the use of high-dose neostigmine, especially if neuromuscular transmission monitoring is not applied. However, use of proper neostigmine reversal guided by neuromuscular transmission monitoring, defined as use of 60mcg/kg or less after recovery of train-of-four count of two, was associated with elimination of increased risk of PRCs.

Fluid administration impacts the course of patients. Both liberal and the most restrictive fluid resuscitation strategies have been associated with respiratory complications (pulmonary oedema or increased dead space during hypovolemia, respectively), whereas moderately restrictive fluid volumes are consistently associated with optimal postoperative outcomes.

Intra-operative haemodynamic instability, expressed as the requirement of high vasopressor doses and transfusion of packed red blood cells, are associated with an increased risk of early tracheal re-intubation.

Some data suggest that very high doses of opioids administered during surgery are associated with increased 30-day readmission rate.

Compared with anaesthesia without an epidural or spinal, neuraxial blockade may reduce postoperative morbidity and mortality in disease-entity based subcohorts. Despite the effects of local anaesthetics on motor function and sympathetic innervation, epidural anaesthesia improves postoperative lung function and reduces PRCs.

A laparoscopic surgical approach may also have beneficial effects on postoperative respiratory outcomes compared

with open surgery.

Prolonged postoperative hypoxemia can occur during recovery from surgery particularly in patients receiving opioids for postoperative analgesia.

Proper monitoring by pulse oximetry surveillance in the PACU as well as subsequently in the ward can decrease the number of ICU transfers and improves patients'™ outcomes by detecting early signs of respiratory complications. Continuous positive airway pressure compared with atmospheric pressure applied through an oronasal mask improved sleep-disordered breathing and ameliorated the respiratory-depressant effects of opioids postoperatively.

Postoperatively, early mobilization improves functional independence. Data demonstrate that early, goal-directed mobilization shortens patients'™ length of stay in the SICU and improves functional mobility at hospital discharge.

This study also aims to observe the patients for any **postoperative complication** till 30 days after completion of surgery. (defined as: postoperative respiratory complications such as presence of any of the stated criteria – Pneumonia, Atelectasis, Pulmonary oedema, Acute Respiratory failure, Requirement of mechanical ventilation > 48 hrs or Reintubation, Surgical site infection – Superficial/ Deep/ Organ space, wound disruption, PE, DVT, progressive renal insufficiency, acute renal failure, UTI, stroke, cardiac arrest, myocardial infarction, systemic sepsis, return to OR, readmission in ICU, readmission in ward after discharge, mortality).

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